

BSESS - F26 - 1A

MA - 216

Discrete Structure

↳ Sir Dr. Tauqueer Ahmed

Assignment . 02

Set Identities Venn Diagram

Week: 04 Lecture: 11, 12

Student: M. Athar Imran

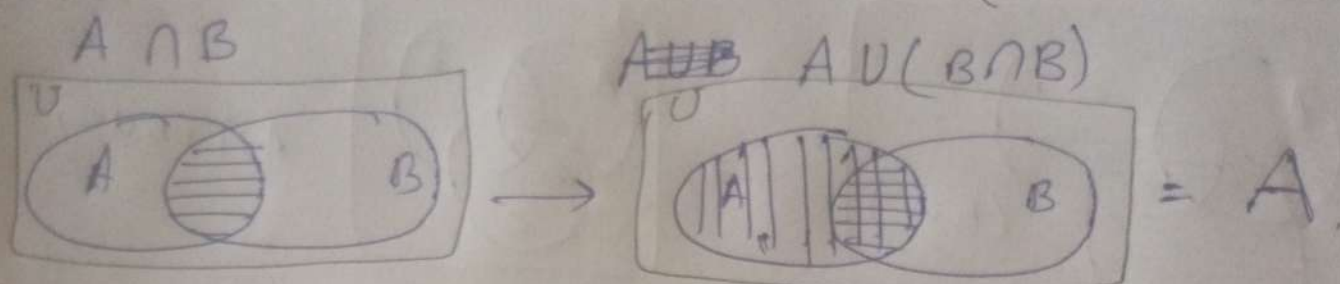
Roll no.: 2604 277

Assign: 08th Oct, 26, thu

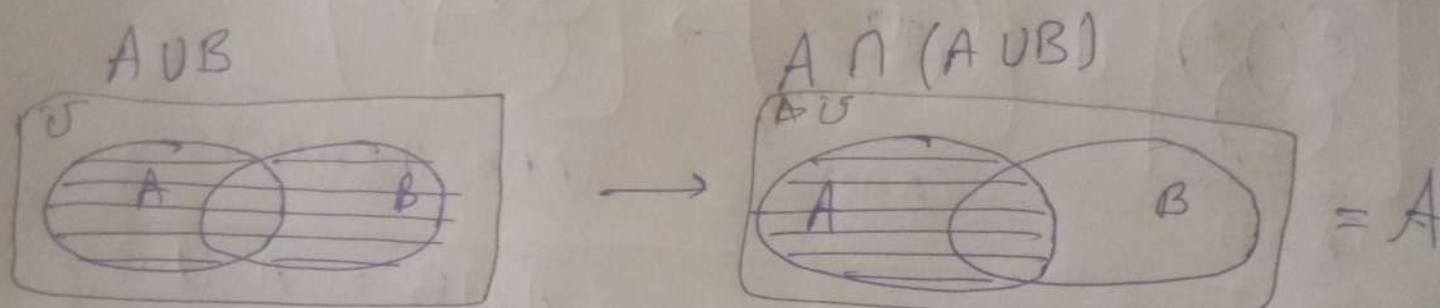
Done: 10th Oct, 26, sat

Due: 15th Oct 26, thu

* Absorption Law: $A \cup (A \cap B) = A$

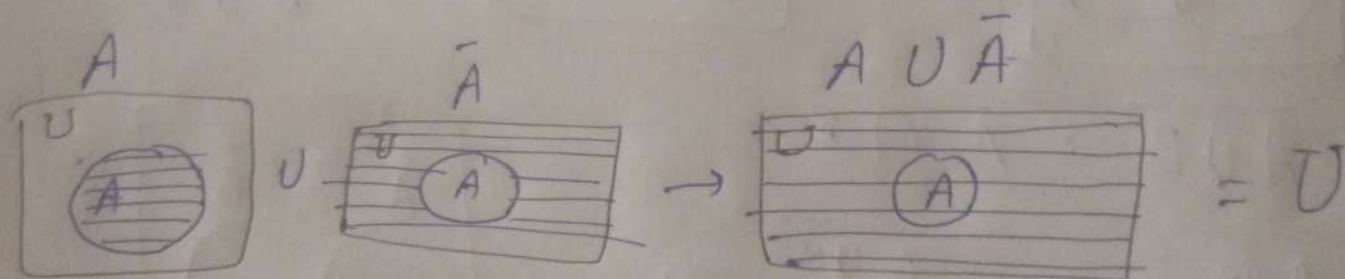


• $A \cap (A \cup B) = A$

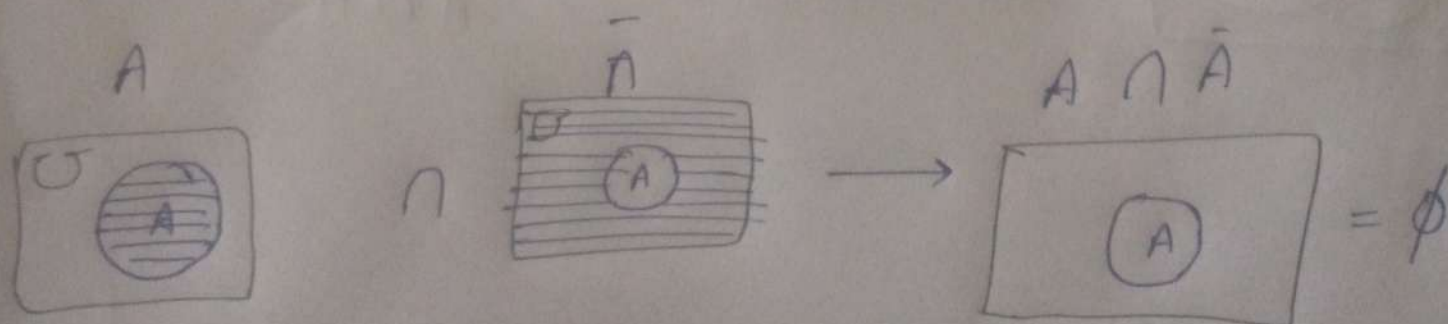


* Complement Law:

• $A \cup \bar{A} = U$



• $A \cap \bar{A} = \emptyset$

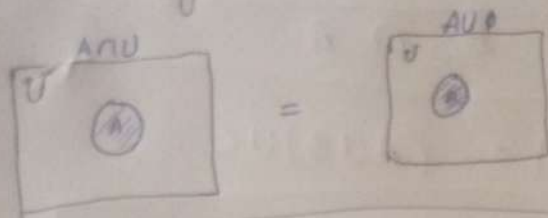


← The End →

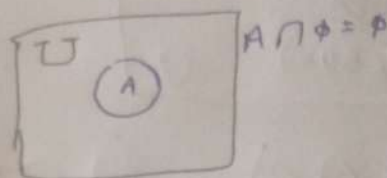
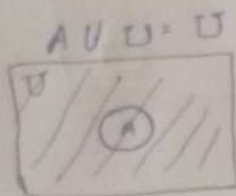
Venn Diagram of Set Identities:-

02

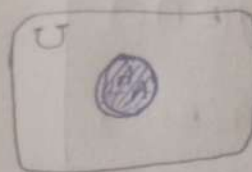
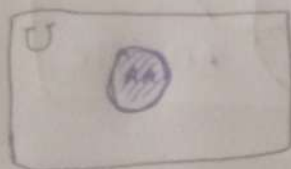
*1. Identity Law: $A \cap U = A = A \cup \phi$



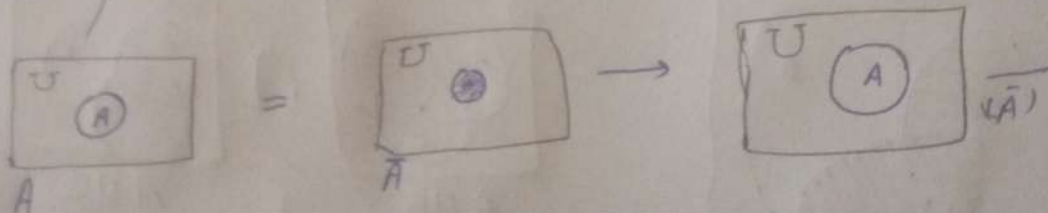
*2. Domination Law: $A \cup U = U$
 $A \cap \phi = \phi$



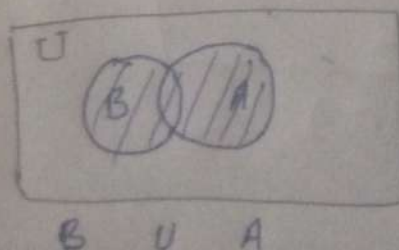
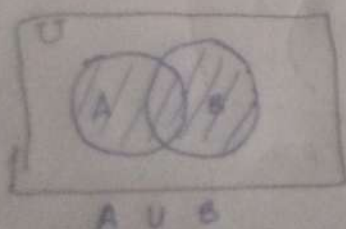
*3. Idempotent Law: $A \cup A = A = A \cap A$



*4. Complementation Law: $\overline{(\bar{A})} = A$

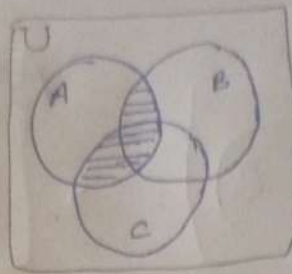
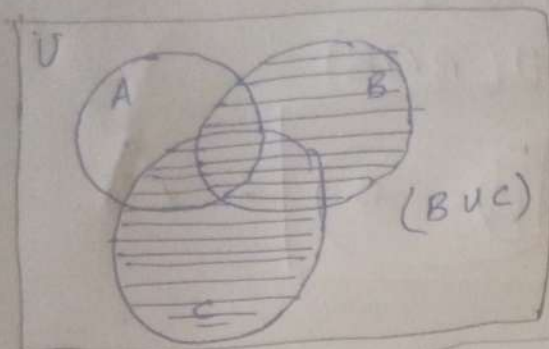


*5. Commutative Law: $A \cup B = B \cup A$

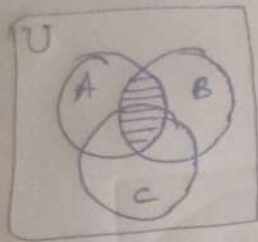


$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

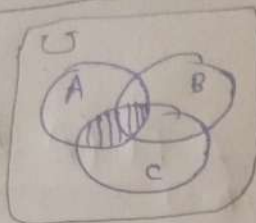
02



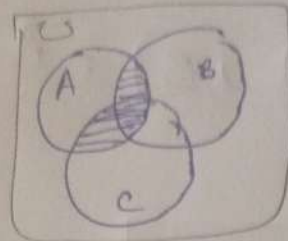
$A \cap (B \cup C)$



$(A \cap B)$

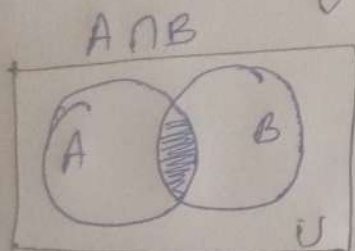


$(A \cap C)$

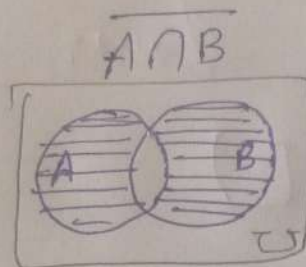
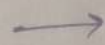


$(A \cap B) \cup (A \cap C)$

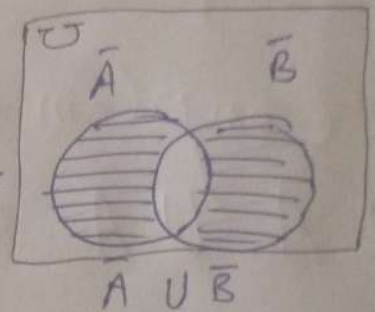
* De Morgan's Law: $\overline{A \cap B} = \bar{A} \cup \bar{B}$



$A \cap B$

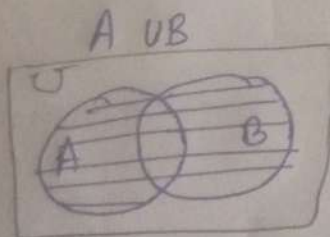


$\overline{A \cap B}$

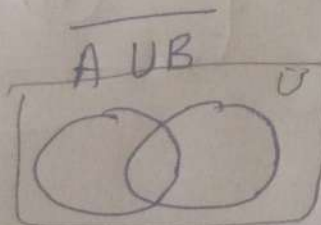


$\bar{A} \cup \bar{B}$

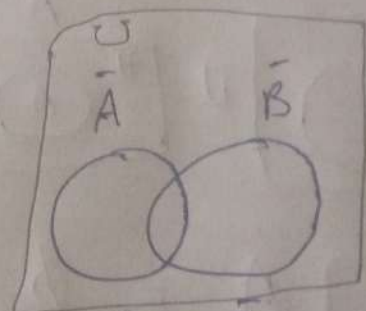
$$\overline{A \cup B} = \bar{A} \cap \bar{B}$$



$A \cup B$

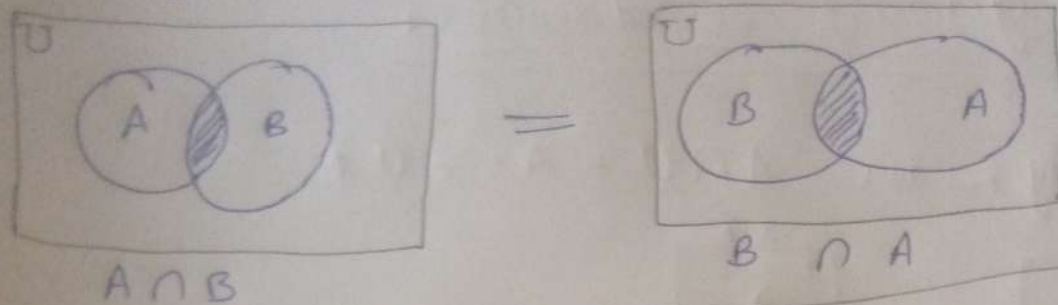


$\overline{A \cup B}$

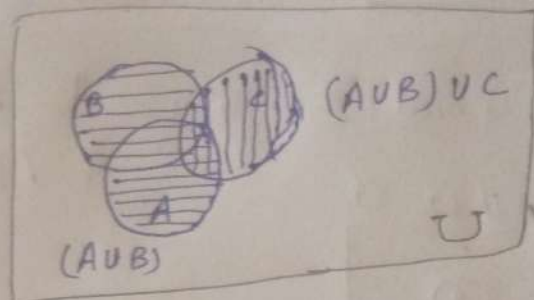
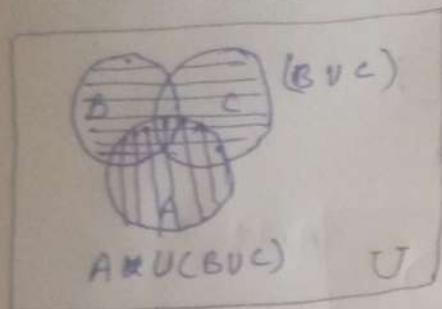


$\bar{A} \cap \bar{B}$

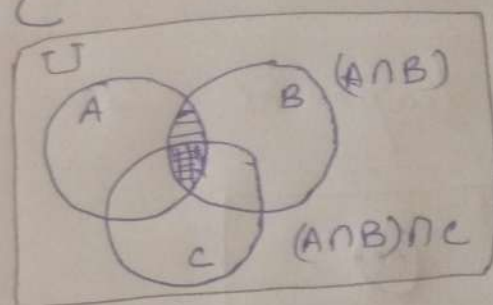
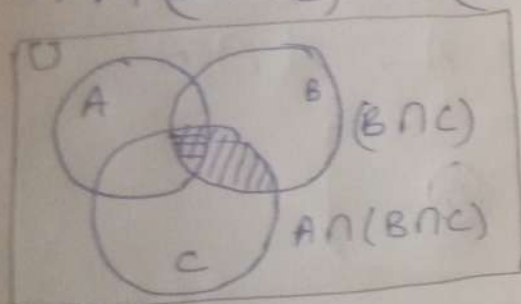
• $A \cap B = B \cap A$



* Associative Law: • $A \cup (B \cap C) = (A \cup B) \cap C$



• $A \cap (B \cap C) = (A \cap B) \cap C$



* Distributive Law: • $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

